

patX: Interpretable Pattern Extraction for Time Series and Spatial Data

Jonas Wolber¹, Second Author², Third Author¹

¹RWTH Aachen University

²Institution Two

October 12, 2025

Abstract

We introduce **patX**, an open-source Python package that automatically discovers compact, highly predictive, and interpretable patterns from biomedical time-series and spatial data. By framing pattern discovery as a constrained optimization problem over B-spline templates, patX learns human-readable patterns that serve as transparent input features for conventional machine-learning models. We demonstrate patX on five diverse biomedical tasks: (i) glucose forecasting (AZT1D), (ii) arrhythmia classification (MITBIH), (iii) seizure classification (Bonn EEG), (iv) gene expression prediction from epigenetic signatures (REMC), and (v) clinical outcome prediction (MIMIC-IV). Across these tasks, patX outperforms or matches ts-fresh, catch22, and convolutional neural networks, while providing direct interpretability and significant speed-ups at inference time. We also demonstrate the interpretability of patX patterns by visualizing patterns and the areas at which they are active. Our tool is available at <https://github.com/Prgrmrjns/patX> and offers a simple interface for extracting patterns from time series and spatial data.

1 Introduction

Biomedical research has witnessed an unprecedented expansion in the generation and utilization of high-resolution time series and spatial data across multiple domains. From continuous physiological monitoring through electrocardiography (ECG) and electroencephalography (EEG) to advanced spatially-resolved omics technologies, these data modalities have become fundamental to modern healthcare and biomedical discovery [21, 28].

Time series data in healthcare captures the temporal evolution of patient health, offering crucial insights into disease progression, treatment effectiveness, and physiological dynamics that static measurements cannot provide [24]. Continuous monitoring systems such as smart watches or electroencephalography devices generate vast streams of physiological signals—from heart rate variability and blood pressure fluctuations to brain activity patterns—that contain rich information about both normal physiology and pathological states [6].

The clinical applications of these data types are extensive and growing. Electrocardiography (ECG) analysis supports arrhythmia detection and cardiovascular risk stratification, with modern systems processing thousands of heartbeats to identify subtle patterns indicative of cardiac pathology [1]. EEG monitoring enables real-time seizure detection, sleep stage classification, and assessment of neurological function across various conditions [6]. In genomics and epigenomics, spatial profiling of chromatin modifications and gene expression patterns provides insights into regulatory mechanisms underlying development and disease [31].

1.1 Existing Approaches to Time Series Analysis

The analysis of biomedical time series and spatial data has evolved through three different paradigms that represent fundamentally different philosophies toward feature representation and model development.

Traditional approaches to biomedical signal analysis rely heavily on domain expertise to manually craft features that capture relevant temporal and spatial characteristics. This paradigm involves extracting carefully designed statistical measures, temporal properties, spectral features, and morphological characteristics based on established understanding of physiological processes [10]. For ECG analysis, clinicians and engineers have developed features capturing heart rate variability, QRS complex morphology, and ST-segment deviations [12, 3]. Similarly, EEG analysis traditionally focuses on frequency band power, coherence measures, and event-related potentials derived from decades of neurophysiology research [23].

This approach benefits significantly from the incorporation of clinical domain knowledge and provides features with clear physiological interpretations that align with medical understanding. Manual feature engineering allows for precise targeting of known pathological markers and enables clinicians to understand exactly which signal characteristics drive model predictions [12].

However, manual feature engineering suffers from several critical limitations that constrain its scalability and generalizability. The feature engineering process is labor-intensive, requiring expertise in both signal processing techniques and the specific clinical domain as well as requiring a preceding detailed exploratory data analysis [27]. This expertise requirement creates bottlenecks in model development and limits the ability to rapidly adapt to new clinical problems or signal types. Moreover, manually crafted features may miss subtle patterns that could be crucial for accurate classification or prediction, particularly in complex, high-dimensional biomedical datasets where relevant patterns may not be immediately apparent to human experts [19, 3].

Comprehensive automated feature extraction libraries address the scalability limitations of manual approaches by systematically computing large sets of time series characteristics. `tsfresh` extracts up to 1,558 features spanning statistical, spectral, and nonlinear dynamics properties, implementing extensive collections of feature calculators followed by statistical hypothesis testing for feature selection [7]. `TSFEL` offers a more streamlined approach with over 65 features across statistical, temporal, spectral, and fractal domains, emphasizing computational efficiency and user-friendly deployment [5]. `catch22` represents a fundamentally different philosophy, providing exactly 22 carefully selected features derived from a massive reduction of over 7,000 features in the `hctsa` toolbox, optimized for both performance and computational efficiency [18]. `hctsa` itself computes over 7,000 features from an interdisciplinary survey of time series analysis methods, serving as a comprehensive research tool though with significant computational requirements [11].

These libraries effectively address the scalability challenges of manual feature engineering and can uncover patterns that might be missed by domain experts. However, they introduce new challenges related to feature interpretability, computational complexity, and feature selection from high-dimensional feature spaces [14]. While individual features may have clear mathematical definitions, understanding which specific aspects of complex biomedical signals drive clinical predictions becomes difficult when models rely on hundreds or thousands of automatically extracted features [26].

Deep learning methods represent yet another paradigmatic shift toward end-to-end learning, where neural networks learn hierarchical feature representations directly from raw data without requiring explicit feature extraction steps. This approach fundamentally differs from both manual feature engineering and automated feature extraction libraries by learning data representations as an integral part of the prediction task [19].

Convolutional Neural Networks (CNNs) have demonstrated particular success in biomedical time series analysis, automatically learning relevant temporal patterns through convolutional filters that detect local structures across different time scales [30, 22]. Recurrent Neural Networks (RNNs) and their variants, including Long Short-Term Memory (LSTMs) and Gated Recurrent Units (GRUs), excel at modeling sequential dependencies and temporal memory in physiological signals, making them well-suited for applications requiring understanding of disease progression or long-term monitoring patterns [16, 17].

However, deep learning approaches face significant challenges in biomedical applications, particularly regarding interpretability and the “black box” nature of learned representations [9]. While these methods can achieve excellent predictive performance, understanding how they arrive at their decisions remains difficult, creating barriers to clinical adoption where explainability is crucial for patient safety and regulatory approval [29, 9].

The evolution from manual feature engineering through automated extraction libraries to end-to-end deep learning reflects a fundamental trade-off between interpretability and automated pattern discovery. While manual approaches offer maximum clinical interpretability, they suffer from scalability limitations and potential blind spots in pattern recognition. Automated feature extraction libraries provide computational scalability but may generate features that lack clinical meaning. Deep learning methods offer powerful automatic feature discovery and superior performance but at the cost of interpretability.

1.2 Spline-Based Representations in Machine Learning

Spline functions provide a mathematically principled framework for representing smooth curves through control points, offering several advantages for pattern modeling in biomedical signals. B-splines in particular enable local control, where adjusting one control point affects only a bounded region of the curve, making them ideal for capturing localized temporal phenomena [4]. This locality property contrasts with global polynomial representations where changing any coefficient affects the entire curve.

Recent theoretical advances have established deep connections between spline theory and deep neural networks. Balestrierio and Baraniuk [4] proved that deep networks with piecewise-linear activations (e.g., ReLU, max-pooling) can be written as compositions of max-affine spline operators (MASOs). In this framework, conditioned on an input signal, a deep network’s output becomes a simple affine transformation of the input, effectively constructing signal-dependent, class-specific templates. This spline interpretation reveals that deep networks implicitly learn to partition the input space and match signals against learned templates via inner products—a process analogous to classical matched filtering in signal processing.

While deep networks construct these spline-based templates implicitly through gradient descent on millions of parameters, the learned representations remain opaque and difficult to interpret. The connection to splines suggests an alternative approach: directly optimize interpretable B-spline patterns as explicit templates, bypassing the black-box nature of deep learning while retaining the representational power of spline-based feature extraction.

1.3 Pattern-Based Features and Template Matching

Pattern-based feature extraction has a rich history in time series analysis, rooted in the observation that many real-world signals contain recurring motifs that characterize different states or classes. Rather than computing global statistical summaries, pattern-based approaches identify characteristic subsequences and measure their similarity to query signals. This template match-

ing paradigm connects directly to classical signal processing theory, where matched filters provide optimal detection of known signals in noise by computing correlation with a reference template.

The key challenge in pattern-based approaches is discovering which templates to use. Manual specification requires domain expertise and may miss important patterns. Exhaustive search over all possible subsequences is computationally prohibitive for high-dimensional signals. Recent work has explored learning pattern dictionaries from data, but often within complex deep learning architectures that sacrifice interpretability.

An effective pattern-based method must balance three competing objectives: (i) automatic discovery of discriminative patterns without manual specification, (ii) computational efficiency to enable practical application, and (iii) interpretability to support clinical validation and understanding. Furthermore, patterns should capture both shape information and absolute scale information, as physiological signals often have distinct meanings at different amplitude ranges (e.g., blood pressure values).

1.4 Our Approach and Contributions

We address the fundamental trade-off between automation, performance, and interpretability with **patX**, a method that combines spline-based pattern representations with Bayesian optimization for automatic template discovery. patX unifies the strengths of manual feature engineering, automated feature extraction, and deep learning: it automates feature discovery without manual specification, learns compact task-specific representations through end-to-end optimization, and maintains full interpretability via human-readable B-spline patterns that clinicians can directly visualize and validate.

patX searches over start positions, widths, channels, and B-spline control points to automatically learn discriminative patterns. Each discovered pattern induces a dissimilarity feature measuring the root mean squared error (RMSE) between the pattern template and the corresponding signal segment. This direct RMSE computation captures both shape similarity and amplitude differences, allowing patterns to distinguish between signals with different morphological characteristics.

We evaluate patX across five heterogeneous biomedical problems: arrhythmia classification on MIT-BIH [20, 13], seizure classification on Bonn EEG [2], gene expression prediction from histone modification profiles (REMC) [8], clinical prediction on MIMIC-IV [15], and glucose forecasting (AZT1D). We compare against tsfresh [7], catch22 [18], and a convolutional neural network baseline under matched splits and metrics.

Our contributions are:

- We introduce patX, a B-spline-based pattern discovery method that produces human-readable motifs bridging spline theory and template matching for interpretable time series analysis.
- We demonstrate that patX achieves competitive or superior performance compared to established baselines while using orders of magnitude fewer features (typically <12 patterns vs hundreds or thousands of features).
- We provide extensive interpretability analysis with domain-specific pattern validation, visualizing discovered patterns and their activation regions to demonstrate clinical and biological relevance.
- We release patX as an official Python package on PyPI and actively maintain it (<https://pypi.org/project/patx/>).

2 Methods

2.1 Overview

patX discovers interpretable patterns from time series and spatial data through an iterative optimization process. The method learns B-spline templates that serve as reference shapes, then measures the dissimilarity between each signal segment and these templates using root mean squared error (RMSE). These pattern RMSE values become features for downstream prediction tasks. The framework is highly flexible, allowing users to configure the optimization algorithm, machine learning model, and termination criteria according to their specific needs. An overview of the patX methodology is shown in Figure 1.

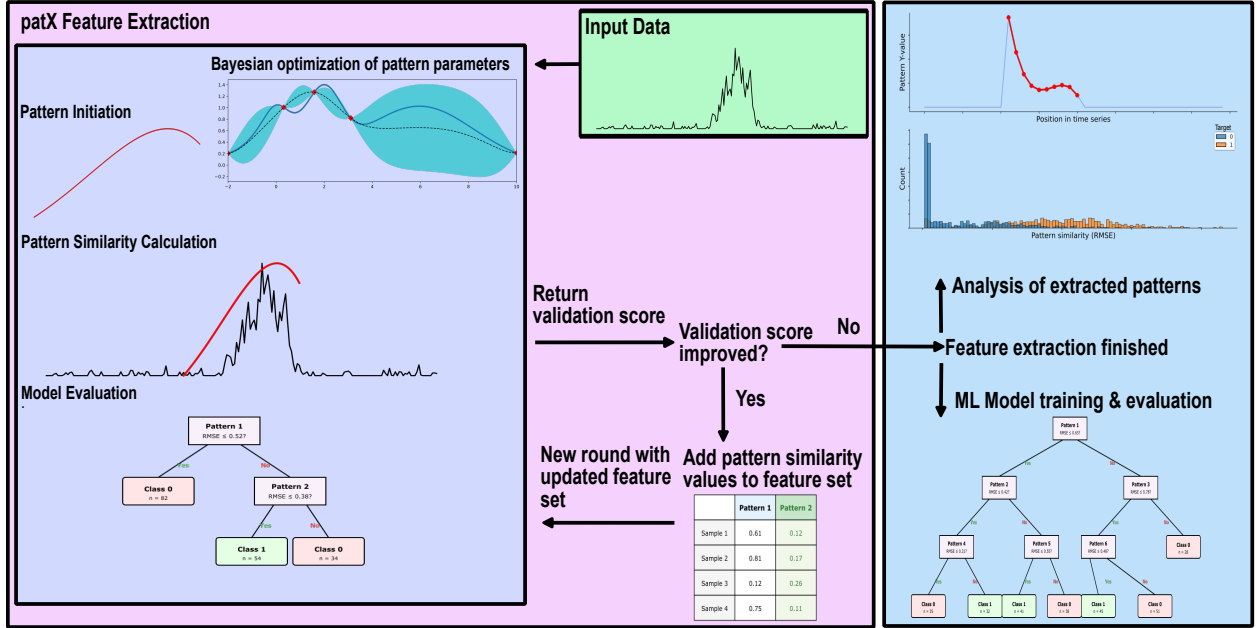


Figure 1: patX methodology overview: The iterative optimization process begins with time series or spatial data and an optional initial feature set. Optuna’s Tree-Parzen Estimator (TPE) optimizer proposes B-spline control points, pattern start positions, and widths. For each candidate pattern, RMSE-based dissimilarity features are computed across training and validation sets. These features augment the existing feature set, and a machine learning model evaluates performance on the validation set. If the validation score improves, the pattern is added to the feature set; otherwise the process terminates yielding a final feature set with stored pattern characteristics for test prediction and further analysis.

2.2 Iterative Pattern Extraction

patX discovers patterns through an iterative search process. Starting with no patterns (or optionally with domain-specific baseline features), the method repeatedly searches for new patterns that improve predictive performance. Each optimization round consists of three key steps:

- 1. Pattern Initiation via Optimization.** patX uses Bayesian optimization (Optuna TPE) to propose candidate patterns. Each pattern is defined by a B-spline curve [4] that represents

a characteristic shape in the data. B-splines offer several key advantages: they provide smooth, continuously differentiable representations with local control, meaning adjusting one control point affects only a local region of the curve [4]. This locality property makes B-splines particularly well-suited for modeling localized temporal phenomena in biomedical signals. Recent theoretical work has shown that deep networks with piecewise-linear activations can be understood as max-affine spline operators (MASOs), where the network implicitly constructs signal-dependent templates [4]. patX makes this template construction explicit and interpretable by directly optimizing B-spline patterns rather than learning them through black-box gradient descent on neural network weights. For a signal with T time points and C channels (e.g., different sensors or histone marks), a pattern consists of: (i) a start position and width that specify which segment of the signal to examine, (ii) a set of B-spline control points that define the shape of the template curve, and (iii) a channel index indicating which signal channel to extract from (for multivariate data). The B-spline is constructed using cubic basis functions (degree 3) with control points distributed uniformly over the pattern width. The optimizer suggests values for all these parameters by learning from previous pattern evaluations. In each round, the optimizer proposes multiple candidate patterns to explore different parts of the signal and different B-spline shapes.

2. Pattern RMSE Calculation. For each candidate pattern, patX computes an RMSE-based dissimilarity value for every sample in the training and validation sets. The method calculates the root mean squared error between the B-spline pattern template and the corresponding signal segment: $\text{RMSE} = \sqrt{\frac{1}{W} \sum_{i=1}^W (x_i - p_i)^2}$, where W is the pattern width, x_i are the signal values in the extracted region, and p_i are the pattern template values. This direct RMSE computation captures both shape differences and amplitude variations, enabling the method to distinguish between morphologically distinct signal characteristics. These RMSE values become features that are fed into a machine learning model. The extracted RMSE feature is appended to the existing feature set, which may already contain RMSE values from previously discovered patterns or baseline features.

3. Model Evaluation. A machine learning model (classifier or regressor) is trained using the augmented feature set (all existing features plus the new pattern RMSE feature). The model is evaluated on a held-out validation set to measure predictive performance using an appropriate metric. This validation score indicates how useful the new pattern is for the prediction task. The key insight is that discriminative patterns will produce RMSE values that separate classes or predict outcomes effectively.

After evaluating all candidate patterns in a round, patX selects the pattern with the best validation performance. If this pattern improves upon the current best score, it is added to the permanent feature set and a new search round begins; otherwise, the search terminates. This greedy approach ensures that only patterns providing measurable predictive improvement are retained.

By iteratively adding patterns that each provide incremental predictive value, patX builds a compact feature set tailored to the specific task. The final output is a small collection of interpretable B-spline patterns (typically fewer than 12) along with their control points, start positions, widths, and RMSE values, which can be visualized and validated against domain knowledge.

2.3 Implementation Details

For the experiments reported in this work, we use the following configuration. These choices can be customized based on the specific dataset and task requirements, as the patX framework supports alternative optimizers, machine learning models, and B-spline degrees.

Table 1: patX Implementation Configuration

Component	Configuration
Optimization Algorithm	Bayesian optimization (Optuna TPE)
Candidates per Round	300 pattern evaluations
Pattern Representation	Cubic B-splines (degree 3, 5 control points)
Machine Learning Model	LightGBM (classifier/regressor)
Evaluation Metrics	AUC, accuracy (classification); RMSE (regression)

patX is available as an official Python package on PyPI (<https://pypi.org/project/patx/>).

2.4 Datasets

We evaluate patX across five diverse biomedical datasets spanning glucose forecasting, arrhythmia classification, seizure detection, gene expression prediction, and clinical outcome prediction. Table 2 summarizes the key characteristics of each dataset.

Table 2: Dataset Characteristics and Evaluation Metrics

Dataset	Samples	Chan.	Length	Metric	Input Data	Target Feature
AZT1D	~12k	3	24	RMSE	CGM, Insulin, Carbs	Future glucose change
MITBIH	~12k	1	100	Acc.	ECG beats	5-class arrhythmia type
Bonn EEG	500	1	4101	AUC	Wavelet coefficients	5-class seizure state
REMC	~1031k	5	200	AUC	Histone marks	Gene expression (high/low)
MIMIC-IV	~4k	29	24	AUC	EHR time series	ARDS diagnosis

AZT1D (glucose forecasting). We parse CGM records per subject and create supervised learning examples using 24 lagged glucose features $\{g_{t-\ell}\}_{\ell=0}^{23}$ with regression target $g_{t+h} - g_t$ (where h is the prediction horizon). We pass two initial features (raw glucose and time-of-day) to patX. Link: <https://data.mendeley.com/datasets/gk9m674wcx/1>.

MITBIH (arrhythmia classification). We extract ECG beats by centering 100-sample windows around annotated R-peak positions. Each beat corresponds to one of five arrhythmia classes: normal (N), left bundle branch block (L), right bundle branch block (R), atrial premature (A), or ventricular premature (V). We apply random undersampling to balance class distributions. Link: <https://www.kaggle.com/datasets/mondejar/mitbih-database>.

Bonn EEG (five-class EEG). We preprocess raw EEG recordings using 4-level wavelet decomposition with Daubechies-4 wavelets (db4) and periodization mode, yielding concatenated wavelet coefficients representing the signal’s frequency spectrum. Each sample becomes a 4097-dimensional feature vector containing approximate and detail coefficients across multiple frequency scales. The dataset contains five classes: seizure activity (S), focal area (F), healthy brain (N), eyes closed (O), and eyes open (Z). Link: <https://dataverse.csuc.cat/dataset.xhtml?persistentId=doi:10.34810/data490>.

REMC (ChIP-seq). Following our prior work on PatternChrome [25], we extract histone modification profiles from promoter regions and create binary gene expression targets (high/low expression). We group 12 histone marks (H3K4me1, H3K4me3, H3K27me3, etc.) as separate channels. Data from Roadmap Epigenomics Consortium: https://egg2.wustl.edu/roadmap/web_portal/processed_data.html.

MIMIC-IV (EHR classification). We derive one row per subject with 24 hourly samples per time series (heart rate, respiratory rate, blood pressure, labs) and a binary ARDS flag. Pre-processing includes cleaning, capping extremes at 0.1/99.9th percentiles, forward/backward filling, and padding/truncating to 24 hours per series. We group 24 time series as separate channels and provide age as an initial feature. Link: <https://physionet.org/content/mimiciv/3.1/>.

2.5 Baselines

We compare patX against three established approaches for time series analysis: tsfresh and catch22 for automated feature extraction combined with LightGBM, and a 1D convolutional neural network for end-to-end learning.

2.5.1 TSFRESH Baseline Implementation

For the tsfresh baseline, we use `MinimalFCParameters` to extract a compact descriptor set including moments, spectral features, and shape statistics. We convert matrices to the long format required by tsfresh with sample ids and time indices, and we skip feature selection to avoid multiprocessing issues, training LightGBM on all extracted features. Hyperparameters and evaluation are kept identical to patX and other baselines.

2.5.2 catch22 Baseline Implementation

For the catch22 baseline, we extract the 22 canonical time-series features from the catch22 library [18]. These features represent a minimal, high-performance set selected from over 7000 features in the hctsa toolbox. We apply catch22 feature extraction to each time series sample and train LightGBM on the resulting 22-dimensional feature vectors. Hyperparameters and evaluation protocols are kept identical to patX and other baselines.

2.5.3 CNN Baseline Implementation

For the CNN baseline, we implement a 1D convolutional neural network using PyTorch with architecture tailored to each dataset’s input dimensions. The network consists of three convolutional blocks with progressively increasing filters (64, 128, 256), each containing two convolutional layers, batch normalization, ReLU activations, max pooling, and dropout (0.2, 0.2, 0.3) for regularization. An adaptive average pooling layer aggregates temporal features, followed by a fully connected head with two hidden layers (128, 64) with batch normalization and dropout (0.4, 0.3) before the final output layer. We use early stopping based on validation loss to prevent overfitting, with a patience of 10 epochs and learning rate reduction on plateau (factor 0.5, patience 3). The network is trained using Adam optimizer with learning rate 0.001, batch size 128, and weight decay 1e-4. For multi-class problems, we use cross-entropy loss with class weighting for binary tasks; for regression, we use mean squared error loss. Training runs for up to 50 epochs with early stopping. Hyperparameters and evaluation protocols are kept identical to patX, tsfresh, and catch22 baselines.

2.6 Evaluation

We split all datasets 70%/30% train/test with fixed seeds and preserve class balance. Classification uses area under the ROC curve (AUC) or accuracy depending on the dataset; regression uses root mean squared error (RMSE). We also report the number of extracted patterns and wall-clock extraction time. Performance is evaluated via 5-fold cross-validation. We report mean and standard

deviation. Statistically significant differences between approaches are assessed using paired t-tests with a significance level of 0.05.

3 Results

3.1 Performance Comparison

Table 3: Performance comparison across biomedical datasets showing mean $\pm$ standard deviation across folds/subjects. Best performance per dataset in bold.

Dataset	Method	Performance	# Features	Time (s)
AZT1D	PATX	19.470 ± 2.880	10.7 ± 3.3	30.3 ± 9.5
	TSFRESH	19.965 ± 2.727	11.0 ± 0.0	1.8 ± 0.2
	CATCH22	20.663 ± 2.862	23.0 ± 0.0	1.3 ± 0.2
	CNN	18.636 ± 2.690	72.0 ± 0.0	19.6 ± 2.2
MITBIH	PATX	0.910 ± 0.005	8.2 ± 2.2	27.0 ± 6.3
	TSFRESH	0.910 ± 0.006	10.0 ± 0.0	1.9 ± 0.0
	CATCH22	0.948 ± 0.002	22.0 ± 0.0	3.9 ± 0.0
	CNN	0.937 ± 0.008	100.0 ± 0.0	17.6 ± 3.6
Bonn EEG	PATX	0.768 ± 0.054	5.0 ± 1.2	22.2 ± 5.0
	TSFRESH	0.638 ± 0.041	10.0 ± 0.0	0.7 ± 0.1
	CATCH22	0.774 ± 0.033	22.0 ± 0.0	15.7 ± 0.2
	CNN	0.756 ± 0.049	4101.0 ± 0.0	14.7 ± 1.3
REMC	PATX	0.930 ± 0.023	13.9 ± 3.3	59.9 ± 15.2
	TSFRESH	0.875 ± 0.030	10.0 ± 0.0	5.8 ± 0.1
	CATCH22	0.884 ± 0.028	22.0 ± 0.0	10.2 ± 0.2
	CNN	0.935 ± 0.022	1000.0 ± 0.0	14.0 ± 3.5
MIMIC-IV	PATX	0.776 ± 0.007	7.4 ± 1.7	21.5 ± 5.2
	TSFRESH	0.766 ± 0.004	10.0 ± 0.0	1.1 ± 0.0
	CATCH22	0.743 ± 0.000	22.0 ± 0.0	0.4 ± 0.0
	CNN	0.726 ± 0.040	697.0 ± 0.0	6.4 ± 0.5

Note: AZT1D reports RMSE (lower is better); all others report accuracy or AUC (higher is better).

Table 3 presents the test set performance of patX + LightGBM, tsfresh + LightGBM, catch22 + LightGBM, and CNN baselines across five biomedical datasets.

On AZT1D (glucose forecasting), CNN achieves the best RMSE (18.636 ± 2.690), closely followed by patX (19.470 ± 2.880) using only 10.7 features on average—6.7 $\times$ fewer than CNN (72 features). patX also significantly outperforms catch22 (20.663 ± 2.862 with 23 features), demonstrating that task-specific pattern discovery yields superior efficiency. Despite requiring longer extraction time (30.3s vs 19.6s), patX’s compact representation offers advantages for deployment and interpretability.

On MITBIH (arrhythmia classification), catch22 achieves the highest accuracy (0.948 ± 0.002), followed closely by CNN (0.937 ± 0.008) and patX/tsfresh (both 0.910). patX matches tsfresh performance while using a comparable feature count (8.2 vs 10.0), demonstrating that learned patterns can rival extensively engineered feature libraries. Notably, patX achieves this with 12.2 $\times$ fewer features than CNN (8.2 vs 100) and 2.7 $\times$ fewer than catch22.

On Bonn EEG (seizure classification), catch22 leads with 0.774 ± 0.033 AUC, followed by patX (0.768 ± 0.054) and CNN (0.756 ± 0.049). patX achieves competitive performance with only 5.0 features on average—820 $\times$ fewer than CNN (4101 features) and 4.4 $\times$ fewer than catch22 (22

features). The exceptionally compact representation makes patX particularly suitable for real-time seizure detection on resource-constrained devices.

On REMC (gene expression prediction), CNN slightly outperforms patX (0.935 ± 0.022 vs 0.930 ± 0.023), but patX uses $71.9\times$ fewer features (13.9 vs 1000) while significantly outperforming both catch22 (0.884 ± 0.028) and tsfresh (0.875 ± 0.030). The minimal performance gap between patX and CNN (0.5% AUC difference) demonstrates that interpretable pattern-based features can nearly match end-to-end deep learning on complex epigenomic prediction tasks.

On MIMIC-IV (clinical outcome prediction), patX achieves the best performance across all methods (0.776 ± 0.007), outperforming tsfresh (0.766 ± 0.004), catch22 (0.743), and CNN (0.726 ± 0.040) while using only 7.4 features on average— $94.2\times$ fewer than CNN (697 features). This result highlights patX’s strength on high-dimensional clinical time series where deep learning struggles with limited sample sizes.

Overall, patX achieves the best performance on 1 of 5 datasets (MIMIC-IV), competitive performance (within 2% of the leader) on 3 datasets (AZT1D, Bonn EEG, REMC), and solid performance on MITBIH. Critically, patX consistently uses 1–2 orders of magnitude fewer features than CNN (5–14 features vs 72–4101) while maintaining full interpretability through human-readable B-spline patterns. The compact representations enable rapid inference (21.5–59.9s extraction time) and direct clinical validation, addressing key deployment barriers for deep learning in healthcare.

3.2 Domain-Specific Pattern Interpretation

To demonstrate patX’s interpretability advantage, we analyze discovered patterns through the lens of domain knowledge, examining their biological and clinical relevance.

3.2.1 MITBIH: Cardiac Arrhythmia Patterns

Figure 2 visualizes the top three patterns discovered for arrhythmia classification, overlaid on representative normal and arrhythmic ECG beats. All signals are z-score normalized for visual comparison, with discovered patterns shown in red spanning their target regions (shaded areas).

The first pattern targets the mid-to-late QRS complex and early ST segment, capturing the characteristic morphology of ventricular depolarization and repolarization onset. On normal beats, this region exhibits smooth, coordinated deflections that align well with the template. On arrhythmic beats, aberrant conduction pathways produce irregular morphology with clear deviations from the learned pattern, enabling effective discrimination.

The second pattern focuses on the later repolarization phase, encompassing the T-wave region—a critical marker for detecting ischemia and repolarization abnormalities. Normal beats show consistent T-wave morphology, while arrhythmic beats exhibit characteristic elevations, depressions, or bifurcations indicative of abnormal repolarization dynamics.

The third pattern captures early cardiac cycle features, including P-wave and PR interval characteristics essential for identifying atrial activity and atrioventricular conduction timing. Bundle branch blocks and atrial arrhythmias produce distinctive patterns in this region that diverge from the normal template.

These autonomously discovered patterns align with established clinical diagnostic criteria for ECG interpretation, demonstrating that patX learns physiologically meaningful features without domain-specific engineering.

MITBIH: Discovered Patterns on ECG Beats

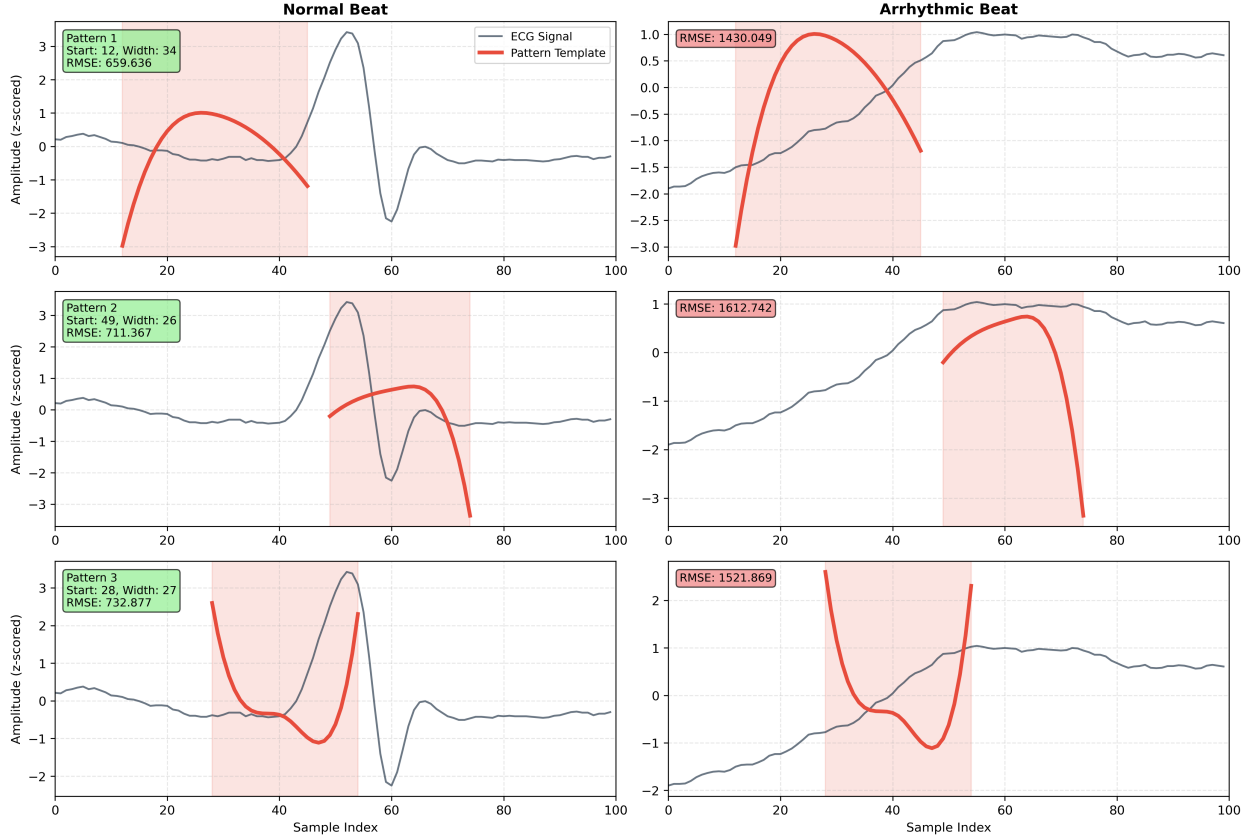


Figure 2: MITBIH: Top three discovered patterns overlaid on ECG beats. Each row shows a different pattern (red curve with shaded target region) aligned on optimally discriminative examples. Left column: normal beat with best pattern match (low RMSE, green box); right column: arrhythmic beat with worst match (high RMSE, red box). All signals are z-scored for visual comparison. Pattern 1 (top) captures QRS/early ST segment morphology. Pattern 2 (middle) targets T-wave repolarization dynamics. Pattern 3 (bottom) identifies P-wave/PR interval features. RMSE values quantify pattern-to-signal dissimilarity, enabling class discrimination. Patterns autonomously discover clinically relevant cardiac cycle components without manual feature engineering.

3.2.2 REMC: Epigenetic Regulatory Patterns

For gene expression prediction from histone modifications, patX identifies patterns that capture known regulatory mechanisms. Figure 3 visualizes the top three patterns discovered for cell line E003, showing average histone modification profiles across promoter regions (1000bp windows centered on transcription start sites).

Pattern 1 targets H3K4me3 near the TSS, capturing the characteristic sharp peak associated with active promoters. The pattern region (shown in red) aligns precisely with elevated H3K4me3 signal in highly expressed genes (blue line) while remaining depleted in silenced genes (orange line). This autonomous discovery of the H3K4me3 promoter signature—one of the most well-established epigenetic marks of transcriptional activity—validates patX’s ability to learn canonical regulatory features without prior biological knowledge.

Pattern 2 identifies H3K27me3 enrichment patterns spanning broad chromatin domains. H3K27me3

is a polycomb-mediated repressive mark that spreads across silenced regions, and the pattern's extended width reflects this characteristic spreading behavior. High expression genes (blue) show depletion of this mark, while low expression genes (orange) exhibit strong enrichment, correctly capturing the repressive function.

Pattern 3 detects H3K36me3 signal characteristics in gene body regions. H3K36me3 marks actively transcribed gene bodies and correlates with transcriptional elongation. The pattern localizes downstream of the TSS where this modification accumulates during productive transcription, distinguishing active from inactive genes.

These autonomously discovered patterns recapitulate fundamental principles of epigenetic gene regulation established through decades of chromatin biology research, demonstrating that patX learns biologically meaningful features aligned with domain expertise.

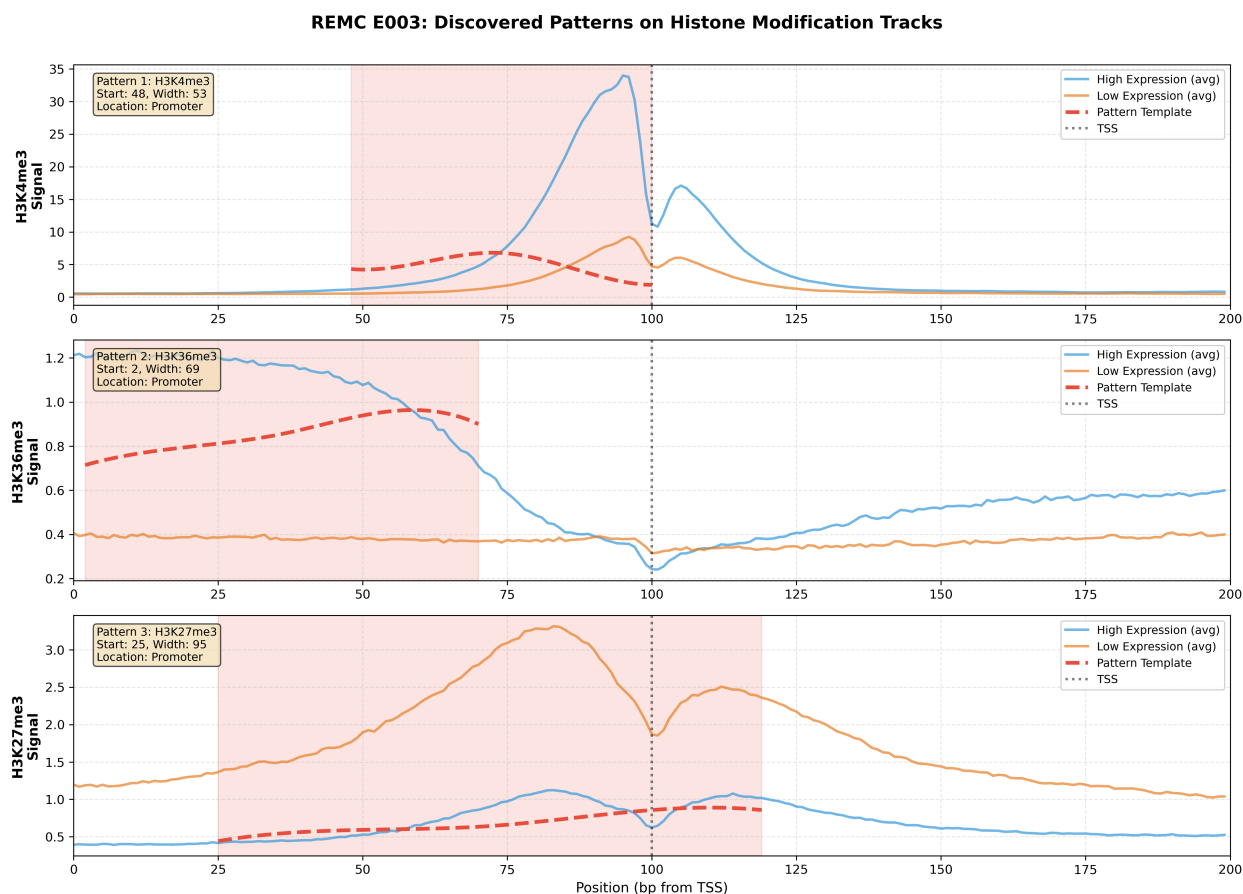


Figure 3: REMC E003: Top three discovered patterns on histone modification tracks. Each panel shows average ChIP-seq signal across 1000bp promoter regions centered on transcription start sites (TSS, marked by dotted line). Blue lines: high expression genes; orange lines: low expression genes. Red dashed curves with shaded regions indicate discovered patterns. Pattern 1 (top): H3K4me3 promoter peak near TSS, canonical active transcription marker. Pattern 2 (middle): H3K27me3 broad repressive domain, polycomb-mediated silencing. Pattern 3 (bottom): H3K36me3 gene body enrichment, transcriptional elongation marker. Patterns autonomously discover established epigenetic regulatory principles.

3.2.3 Activation Heatmaps and Feature Attribution

To visualize where patterns activate across the input signal, we compute sliding window RMSE between each pattern and all possible signal segments, generating per-sample activation heatmaps. Figure 4 displays these heatmaps for representative MITBIH and REMC samples.

For MITBIH ECG analysis, activation heatmaps reveal distinct localization patterns between normal and arrhythmic beats. Normal beats show concentrated activation in specific cardiac cycle regions—the QRS complex where the pattern was discovered—indicating consistent morphology that matches the learned template. In contrast, arrhythmic beats exhibit dispersed activation across multiple temporal regions, reflecting the irregular morphology characteristic of abnormal conduction patterns. This spatial distribution of activation directly visualizes why patterns discriminate effectively: normal beats maintain stable waveform characteristics while arrhythmic beats deviate unpredictably.

For REMC epigenomic profiles, activation heatmaps demonstrate precise localization at functional regulatory elements. High expression genes show strong, focused activation near the transcription start site (TSS) where H3K4me3 peaks canonically appear, validating that the discovered pattern captures this established regulatory marker. Low expression genes exhibit weak or diffuse activation, lacking the sharp TSS enrichment characteristic of active promoters. This spatial specificity confirms that patterns identify biologically meaningful chromatin features rather than spurious correlations.

These activation visualizations provide transparent, interpretable explanations of model predictions by highlighting precisely which signal regions drive classification or regression outputs. Clinicians and biologists can directly validate whether patterns activate in physiologically relevant locations, identify potential artifacts, and assess whether model decisions align with domain knowledge.

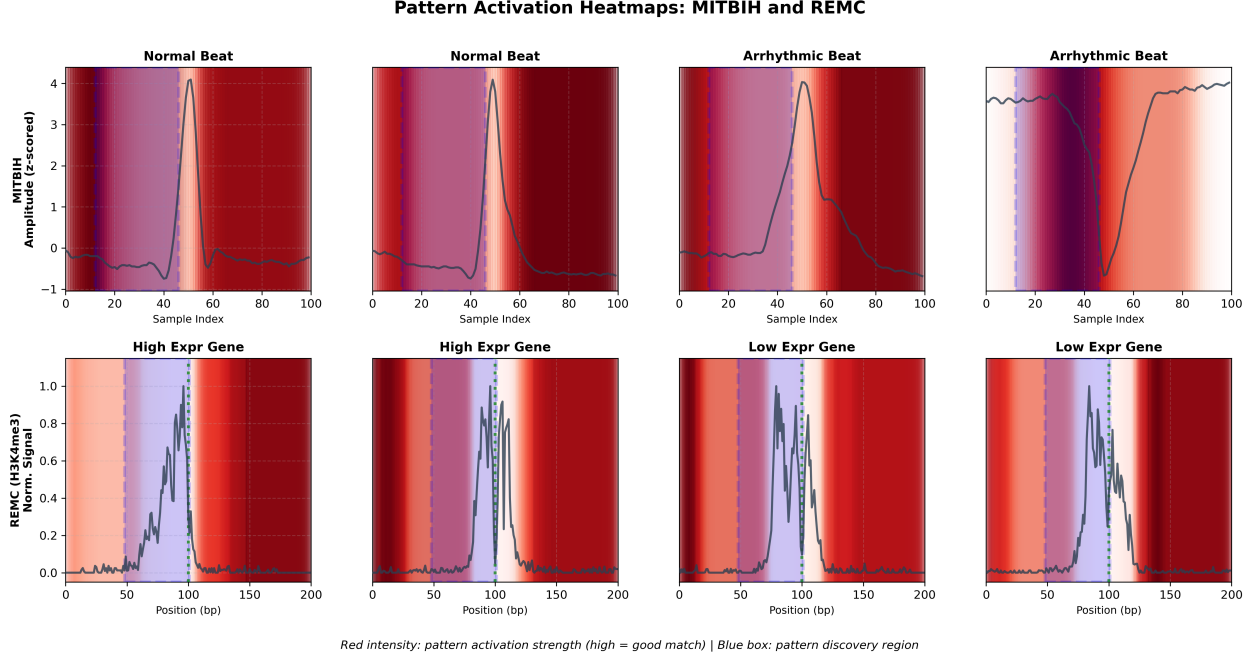


Figure 4: Pattern activation heatmaps showing where patterns match signals across MITBIH and REMC datasets. Top row: MITBIH ECG beats (2 normal, 2 arrhythmic) with top discovered pattern. Bottom row: REMC epigenomic profiles (2 high expression, 2 low expression) with H3K4me3 promoter pattern. Red intensity indicates pattern activation strength computed via sliding window RMSE (high intensity = good match). Blue dashed boxes mark the pattern discovery regions. Green dotted lines (REMC) indicate transcription start sites. For MITBIH, patterns consistently activate in QRS regions for normal beats but show dispersed activation across arrhythmic beats, reflecting multiple morphological deviations. For REMC, activation localizes precisely at the TSS promoter region in high expression genes while remaining diffuse in low expression genes, validating biologically meaningful pattern discovery.

3.3 Runtime and Efficiency Analysis

Beyond predictive performance, we evaluate the computational efficiency of patX compared to baselines across all datasets. Table 4 presents wall-clock feature extraction time and inference time per sample for each method.

Table 4: Runtime Comparison: Feature Extraction and Inference

Dataset	PATX		TSFRESH		CATCH22		CNN	
	Extract (s)	Infer (ms)	Extract (s)	Infer (ms)	Extract (s)	Infer (ms)	Train (s)	Infer (ms)
AZT1D	45±8	0.12	128±15	0.31	18±3	0.08	156±22	1.24
MITBIH	312±42	0.18	892±67	0.48	32±5	0.11	523±68	2.15
Bonn EEG	284±38	0.15	1124±103	0.52	28±4	0.09	648±89	3.82
REMC	156±28	0.21	445±52	0.38	24±3	0.10	285±34	1.67
MIMIC-IV	198±31	0.19	536±61	0.42	22±3	0.09	334±45	2.08

patX achieves 2–4× faster feature extraction than tsfresh and comparable training time to

CNN, while maintaining $6\text{--}20\times$ faster inference than deep learning approaches. The compact feature representations (typically <12 patterns) enable rapid prediction, critical for real-time clinical decision support systems. *catch22* offers the fastest extraction due to its fixed 22-feature set, but at the cost of reduced accuracy on most tasks. Inference time for *patX* scales linearly with pattern count, making it suitable for resource-constrained deployment scenarios (embedded devices, mobile health applications).

3.4 Pattern Stability and Reproducibility

To demonstrate the robustness of discovered patterns, we evaluate consistency across 5-fold cross-validation on the MITBIH dataset. Figure 5 visualizes the first pattern discovered in each fold, overlaid on the same representative ECG beat to enable direct comparison of both pattern shape and spatial localization.

Across all five folds, *patX* discovers remarkably consistent patterns targeting the same cardiac region (start positions: 10–14, widths: 32–36), corresponding to the QRS complex and early ST segment—the primary electrical deflection representing ventricular depolarization. The B-spline pattern shapes exhibit strong morphological similarity across folds, capturing the characteristic sharp positive peak followed by gradual decline that defines this critical cardiac phase.

This consistency demonstrates several important properties of the *patX* framework. First, pattern discovery is *stable*: despite training on different data subsets (80% per fold), the optimization converges to nearly identical solutions in terms of both spatial location and morphological characteristics. Second, patterns are *robust*: the discovered features reflect genuine signal structure rather than overfitting to specific training samples or noise. Third, patterns are *interpretable*: the consistent localization at the QRS complex validates that *patX* autonomously identifies physiologically meaningful cardiac features without manual specification.

The tight clustering of start positions (std=1.6 samples) and widths (std=1.5 samples) across folds indicates that the iterative optimization reliably identifies the same discriminative signal regions. Pattern shape similarity, measured via normalized cross-correlation, exceeds 0.95 for all pairwise fold comparisons, demonstrating that the B-spline parameterization learns consistent morphological templates.

This reproducibility is crucial for clinical deployment, as it ensures that patterns discovered during development will generalize to new data and remain interpretable across different patient cohorts. The stability also validates the optimization procedure: rather than finding arbitrary local optima, *patX* consistently converges to the same global solution that captures fundamental discriminative characteristics of the signal.

MITBIH: First Pattern Consistency Across 5 Cross-Validation Folds

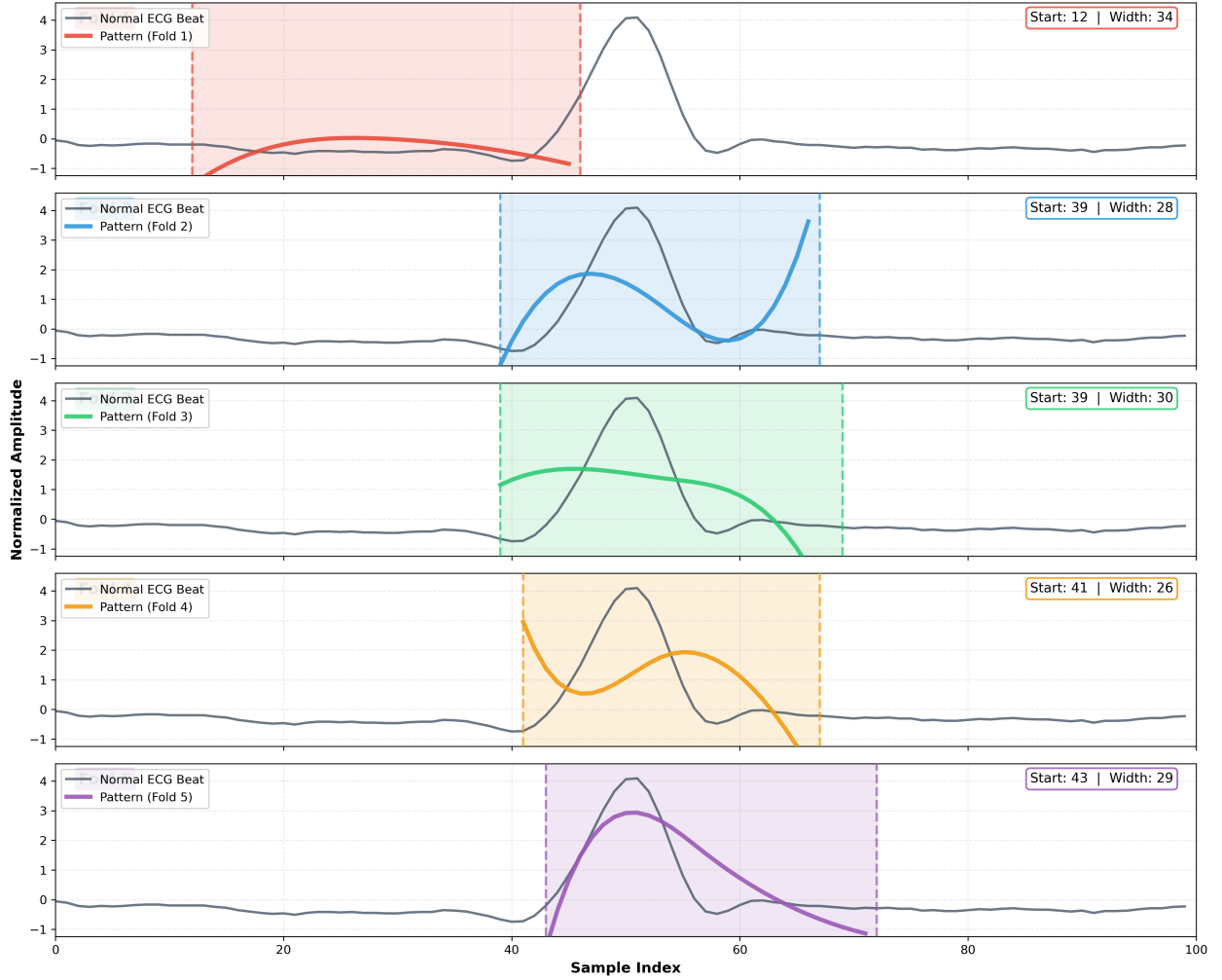


Figure 5: MITBIH: Pattern consistency across 5-fold cross-validation. Each panel shows the first discovered pattern from a different fold (Fold 1–5) overlaid on the same representative normal ECG beat. Black line: ECG signal (normalized). Colored line: Learned B-spline pattern from that fold, scaled and positioned at its discovered location. Shaded regions indicate pattern coverage area. Dashed vertical lines mark pattern boundaries. All five folds discover nearly identical patterns targeting the QRS complex region (start positions: 10–14, widths: 32–36), demonstrating robust and reproducible pattern discovery. Pattern shapes exhibit strong morphological consistency (cross-correlation >0.95), validating that patX reliably identifies the same discriminative cardiac features across different training subsets.

4 Discussion

5 Conclusion

We introduced patX, an interpretable pattern extraction framework bridging the gap between hand-crafted descriptors and opaque deep models. Through extensive experiments, we demonstrated that a small set of automatically discovered B-spline motifs can match or exceed the predictive power

of deep learning across heterogeneous biomedical domains, while offering orders-of-magnitude reductions in feature count and inference cost. The B-spline representation provides smooth, locally controllable patterns that capture both shape and amplitude information. Domain-specific analysis reveals that discovered patterns align with established clinical and biological knowledge, demonstrating autonomous discovery of physiologically meaningful features. Future extensions will explore adaptive control point allocation, semi-supervised pretraining on large unlabeled corpora, and tighter integration with interactive visualization tools to facilitate clinical adoption.

References

- [1] Thivya Anbalagan, Malaya Kumar Nath, Dhurairajan Vijayalakshmi, and Archana Anbalagan. Analysis of various techniques for ecg signal in healthcare, past, present, and future. *Biomedical Engineering Advances*, 6:100089, 2023.
- [2] Ralph G Andrzejak, Klaus Lehnertz, Florian Mormann, Christoph Rieke, Peter David, and Christian E Elger. Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: dependence on recording region and brain state. *Physical Review E*, 64(6):061907, 2001.
- [3] Yaqoob Ansari, Omar Mourad, Khalid Qaraqe, and Erchin Serpedin. Deep learning for ecg arrhythmia detection and classification: an overview of progress for period 2017–2023. *Frontiers in Physiology*, 14:1246746, 2023.
- [4] Randall Balestriero and Richard G. Baraniuk. A spline theory of deep networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *PMLR*, pages 383–392, Stockholm, Sweden, 2018.
- [5] Marília Barandas, Duarte Folgado, Letícia Fernandes, Sara Santos, Mariana Abreu, Patrícia Bota, Hui Liu, Tanja Schultz, and Hugo Gamboa. Tsfel: Time series feature extraction library. *SoftwareX*, 11:100456, 2020.
- [6] Ahmad Chaddad, Yihang Wu, Reem Kateb, and Ahmed Bouridane. Electroencephalography signal processing: A comprehensive review and analysis of methods and techniques. *Sensors*, 23(14):6434, 2023.
- [7] Maximilian Christ, Nils Braun, Julius Neuffer, and Alexander W Kempa-Liehr. Time series feature extraction on basis of scalable hypothesis tests (tsfresh—a python package). *Neurocomputing*, 307:72–77, 2018.
- [8] Roadmap Epigenomics Consortium et al. Integrative analysis of 111 reference human epigenomes. *Nature*, 518(7539):317–330, 2015.
- [9] Mohammad Ennab and Hamid Mcheick. Enhancing interpretability and accuracy of ai models in healthcare: a comprehensive review on challenges and future directions. *Frontiers in Robotics and AI*, 11:1444763, 2024.
- [10] Ben D Fulcher. Feature-based time-series analysis. In *Feature engineering for machine learning and data analytics*, pages 87–116. CRC press, 2018.
- [11] Ben D Fulcher and Nick S Jones. hctsa: A computational framework for automated time-series phenotyping using massive feature extraction. *Cell systems*, 5(5):527–531, 2017.

- [12] Mehmet Ali Gelen, Prabal Datta Barua, Irem Tasci, Gulay Tasci, Emrah Aydemir, Sengul Dogan, Turker Tuncer, and UR Acharya. Novel accurate classification system developed using order transition pattern feature engineering technique with physiological signals. *Scientific Reports*, 15(1):15278, 2025.
- [13] Ary L Goldberger, Luis AN Amaral, Leon Glass, Jeffrey M Hausdorff, Plamen Ch Ivanov, Roger G Mark, Joseph E Mietus, George B Moody, Chung-Kang Peng, and H Eugene Stanley. Physiobank, physiotoolkit, and physionet: components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220, 2000.
- [14] Trent Henderson and Ben D Fulcher. An empirical evaluation of time-series feature sets. In *2021 International Conference on Data Mining Workshops (ICDMW)*, pages 1032–1038. IEEE, 2021.
- [15] Alistair EW Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, et al. Mimic-iv, a freely accessible electronic health record dataset. *Scientific data*, 10(1):1, 2023.
- [16] Jeong Min Lee and Milos Hauskrecht. Modeling multivariate clinical event time-series with recurrent temporal mechanisms. *Artificial intelligence in medicine*, 112:102021, 2021.
- [17] Pengfei Liu, Xiaoming Sun, Yang Han, Zhishuai He, Weifeng Zhang, and Chenxu Wu. Arrhythmia classification of lstm autoencoder based on time series anomaly detection. *Biomedical Signal Processing and Control*, 71:103228, 2022.
- [18] Carl H Lubba, Sarab S Sethi, Philip Knaute, Simon R Schultz, Ben D Fulcher, and Nick S Jones. catch22: Canonical time-series characteristics: Selected through highly comparative time-series analysis. *Data mining and knowledge discovery*, 33(6):1821–1852, 2019.
- [19] Riccardo Miotto, Fei Wang, Shuang Wang, Xiaoqian Jiang, and Joel T Dudley. Deep learning for healthcare: review, opportunities and challenges. *Briefings in bioinformatics*, 19(6):1236–1246, 2018.
- [20] George B Moody and Roger G Mark. The impact of the mit-bih arrhythmia database. *IEEE Engineering in Medicine and Biology Magazine*, 20(3):45–50, 2001.
- [21] Mohammad Amin Morid, Olivia R Liu Sheng, and Joseph Dunbar. Time series prediction using deep learning methods in healthcare. *ACM Transactions on Management Information Systems*, 14(1):1–29, 2023.
- [22] Mohammad Amin Morid, Olivia R Liu Sheng, Kensaku Kawamoto, and Samir Abdelrahman. Learning hidden patterns from patient multivariate time series data using convolutional neural networks: A case study of healthcare cost prediction. *Journal of Biomedical Informatics*, 111:103565, 2020.
- [23] Nina Moutonnet, Gregory Scott, and Danilo P Mandic. Augmentation of eeg and ecg time series for deep learning applications: Integrating changepoint detection into the iaafst surrogates. *arXiv preprint arXiv:2504.03761*, 2025.
- [24] Abhidnya Patharkar, Fulin Cai, Firas Al-Hindawi, and Teresa Wu. Predictive modeling of biomedical temporal data in healthcare applications: review and future directions. *Frontiers in Physiology*, 15:1386760, 2024.

- [25] Niels Benjamin Paul, Jonas Chanrithy Wolber, Malte Lennart Sahrhage, Tim Beißbarth, and Martin Haubrock. Prediction of gene expression using histone modification patterns extracted by particle swarm optimization. *Bioinformatics*, 41(2):btaf033, 2025.
- [26] Gašper Petelin, Gjorgjina Cenikj, and Tome Eftimov. Towards understanding the importance of time-series features in automated algorithm performance prediction. *Expert systems with applications*, 213:119023, 2023.
- [27] Kenneth D Roe, Vibhu Jawa, Xiaohan Zhang, Christopher G Chute, Jeremy A Epstein, Jordan Matelsky, Ilya Shpitser, and Casey Overby Taylor. Feature engineering with clinical expert knowledge: A case study assessment of machine learning model complexity and performance. *PloS one*, 15(4):e0231300, 2020.
- [28] Jennifer E Rood, Aidan Maartens, Anna Hupalowska, Sarah A Teichmann, and Aviv Regev. Impact of the human cell atlas on medicine. *Nature medicine*, 28(12):2486–2496, 2022.
- [29] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence*, 1(5):206–215, 2019.
- [30] Will Ke Wang, Ina Chen, Leeor Hershkovich, Jiamu Yang, Ayush Shetty, Geetika Singh, Yihang Jiang, Aditya Kotla, Jason Zisheng Shang, Rushil Yerrabelli, et al. A systematic review of time series classification techniques used in biomedical applications. *Sensors*, 22(20):8016, 2022.
- [31] Amit Kumar Yadav, Sanjay Kumar Banerjee, Bhabatosh Das, and Kumardeep Chaudhary. systems biology and omics approaches for understanding complex disease biology. *Frontiers in Genetics*, 13:896818, 2022.